

The empirical recurrence for OEIS A230616, recovered from its tail

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Abstract

OEIS A230616 records “Empirical recurrence of order 59” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 344$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 64$. Its last coefficient is $c_{59} = 243226348560 \neq 0$, so no further tail satisfies anything shorter; any order-59 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A230616 is “Number of $4 \times n$ 0..2 arrays $x(i,j)$ with each element horizontally or antidiagonally next to at least one element with value $(x(i,j)+1) \bmod 3$, and upper left element zero.”. It has offset 1 and begins

0, 0, 0, 348, 7072, 149664, 3023532, 61203362, ...

The entry states:

Empirical recurrence of order 59 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 05:55:47, revision 7) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 344$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 344$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 59: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 59.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 688$ exact terms from $n = 5$ on, it returns order 59 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{59} c_i a(n-i),$$

c_1	47	c_{16}	241002098768	c_{31}	891828734879047	c_{46}	−1001855278935346
c_2	−805	c_{17}	−828444447978	c_{32}	−559980815671284	c_{47}	−560234628918044
c_3	6134	c_{18}	1601383169250	c_{33}	365171908352527	c_{48}	149941859488335
c_4	−11703	c_{19}	4268376503580	c_{34}	115551020445294	c_{49}	371846000400507
c_5	−133134	c_{20}	−29762981664352	c_{35}	−797616450883504	c_{50}	149831774200015
c_6	852361	c_{21}	74662880926668	c_{36}	44763510572989	c_{51}	−48842569334612
c_7	−131846	c_{22}	−137668041463823	c_{37}	632188769687799	c_{52}	−69019556913028
c_8	−13935436	c_{23}	267635015780653	c_{38}	1097028002288401	c_{53}	−55415154446957
c_9	40531059	c_{24}	−486953640840146	c_{39}	−685387482531830	c_{54}	3950699148060
c_{10}	−18875012	c_{25}	697816813071623	c_{40}	−1213845145098504	c_{55}	30167869203138
c_{11}	267770572	c_{26}	−882962639331684	c_{41}	−1097335261499405	c_{56}	3857691044620
c_{12}	−1809864504	c_{27}	1127187367207556	c_{42}	407365439681084	c_{57}	−5284500914784
c_{13}	−1457822522	c_{28}	−1268356796740560	c_{43}	1615415828060202	c_{58}	−706181156592
c_{14}	36061038772	c_{29}	1154631700234418	c_{44}	970952183866886	c_{59}	243226348560
c_{15}	−112212009782	c_{30}	−1033227324002487	c_{45}	−242787019279260		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 64$, and it is the recurrence OEIS A230616 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{59} - c_1 x^{59-1} - \dots - c_{59}$. Its constant term is $-c_{59} = -243226348560$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 59 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 59, so $\deg q = 59$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 59, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 59, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 243226348560.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-59 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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